Aroudra Thakur

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Summary

Computer Science Honors student (B.S., expected May 2027) with three internships building **Python and TypeScript tools, workflow automation, and LLM-powered applications**. Experienced with **AI-assisted development tools** (Cursor, Claude Code), REST APIs, Git, testing and debugging, dashboards, and working with business stakeholders.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C/C++
AI & Automation: Large Language Models (LLMs), LangGraph, LangChain, RAG, Embeddings, Prompt Engineering, Workflow Automation, LLM Evaluation; AI coding tools: Cursor, Claude Code, Kiro, GPT Codex
APIs & Data: REST APIs, GraphQL, FastAPI, Django, Node.js, PostgreSQL, Supabase, Snowflake, Redis, Celery, dbt
Frontend, Cloud & Tools: React, Next.js, AWS (S3, Cognito), Docker, Git, GitHub Actions, CI/CD, Pytest, Vitest, Playwright, Debugging, Code Review, Technical Documentation

Experience

Artificial Intelligence Intern *Jun 2026 – Aug 2026*
HF Sinclair *Dallas, TX*

- Built a **Python/LangGraph multi-agent tool** on Snowflake Cortex Search for an enterprise contract-analysis workflow, integrating retrieval, routing, tool execution, structured outputs, validation, and error handling.
- Created **ObsFlow**, a Snowflake-backed observability framework capturing agent traces, latency, citation provenance, and failures to debug and verify AI-generated outputs; built an **80-question evaluation suite** and cut measured token use approximately **70%** (20K to 6K per request).
- Traced a citation failure to an incompatible Snowflake function, fixed it, documented architecture and debugging steps, and presented results to engineers and business stakeholders.

AI Engineering Intern *Dec 2025 – May 2026*
Ozcorp Scientific LLC *Remote (Guaynabo, PR)*

- Developed a full-stack **AI research agent** (Python, FastAPI, React, REST APIs, embeddings) that connects data from multiple external sources, resolves entities, and generates reports, cutting manual research work by **80%+**.
- Added source and link verification, LLM-based relevance scoring, and fallback handling to check AI outputs; improved result quality by **30%**, reduced irrelevant results by **35%**, and cut latency by **50%**.

Software Engineering Intern *Aug 2024 – Dec 2024*
Hotspring *Remote (Barrington, NH)*

- Deployed production **React/TypeScript** and **Django/GraphQL** features (PostgreSQL, AWS S3) and automated media workflows with an email-parsing service and Celery workers, cutting manual file search by **95%+**.

Director of Technology *Jul 2026 – Nov 2026*
HackUTA *Arlington, TX*

- Led **5 engineers** building a React/TypeScript/Convex platform for 1,000+ applications; scoped requirements with organizers, reviewed pull requests, and managed deployments.
- Implemented email OTP authentication, RBAC, rate limiting, and audit logs; set up **GitHub Actions CI/CD** with Vitest, Playwright, and 90% coverage gates.

Project Manager, Silo *Jan 2026 – Present*
ACM (Create Committee), University of Texas at Arlington *Arlington, TX*

- Lead **7 engineers** building Silo, an LLM prompt-evaluation tool; architected the drift-testing pipeline (baseline vs. candidate comparison, semantic scoring, CI integration) and published an npm CLI (silo-drift-cli).

Projects

Nixo Slackbot | *TypeScript, Next.js, OpenAI, Supabase, Socket.IO*
- Automated support-message triage for Forward-Deployed Engineers: classifies Slack messages with OpenAI, groups them into tickets, and updates a live dashboard, with sequential processing for consistent ticket state.

AutoResume | *Next.js, FastAPI, pgvector, Celery, AWS Cognito*
- Built a resume-generation tool using pgvector semantic matching, Celery/Redis document workflows, and deterministic LaTeX rendering; 80%+ backend test coverage.

Education

University of Texas at Arlington *Expected May 2027*
B.S. in Computer Science (Honors) | **GPA: 3.86/4.00** | Minor: Mathematics | Maverick Academic Scholarship
Alpha Lambda Delta | Honors College | President's List | Dean's List | Freshman Distinction Roll

Publications

Coauthor, "Human Pose Estimation in 2D and 3D: A Survey," *Journal of Experimental and Theoretical Analyses*, 2026. Author, "Assistive Technologies for Learning Disabilities: A Systematic Review of Trends and Impact," ASEE-GSW, 2025.